NIHVA

BLDCW & BLDCX Barrier Gate

Instruction Manual

(First Edition)

(Take The Object As The Standard)

1.Product Outline

The FB-3XX BLDC barrier gate series, which researched developed and manufactured by our company, adopts international advanced technology, mechanical and electrical integrated design. Planetary gear drive, high torque, un-blocked, oil tight, no noise, automatically unlocked when the power is off, with 10 million operation times. The motor uses the DC 24V permanent magnet synchronous motor (PMSM motor), adjust to the global voltage, high-speed operation and rapid response, it will not overheat after 24 hours, frequent operation, speed time can be adjusted based on the vehicle flow and arm length, safe and reliable. The series servo barrier gate is free-debugging and free-maintenance, achieve the fully automated and intelligent, and the barrier arm open/close is fast and smooth, easy to use and safe.

2. Operation Time and Boom Length Configuration Parameter

Boom Type(meter)	Operation Time(seconds)	Boom Height(meter)
Fast speed boom≤3	1	0.83
Common straight boom≤4.5	2.5	0.83
Common straight boom≤6	3.5	0.83
80*45mm 90°folding boom≤3	1	0.83
80*45mm 90°folding boom≤3.5	2.5	0.83
80*45mm 90°folding boom≤5	3.5	0.83
80*45mm 180°folding boom≤3	1	0.83
80*45mm 180°folding boom≤4	2.5	0.83
80*45mm 180°folding boom≤5	3.5	0.83
100*45mm 2 fence boom≤3.5	2.5	0.83m, small vertical boom is 0.8m, landing boom is 0.9m
100*45mm 2 fence boom≤5	4	0.83m, small vertical boom is 0.8m, landing boom is 0.9m
100*45mm 3 fence boom≤3	3	0.83m, small vertical boom is 1.5m, landing boom is 1.6m
100*45mm 3 fence boom≤4	4	0.83m, small vertical boom is 1.5m, landing boom is 1.6m

Notice: the “Boom Height ” means height from the boom holding point to the ground.

3. Functions and Features

1. The gate automatically open when the power is off, no clutch design, no need human intervention. When the power is on, the gate automatically reset, and the gate returns to being closed, truly unattended, fire safety standards. If there is special need, manual operation is required when the power if off, the operating arm can open and close manually. Can also set up the control board DIP switch, which will ensure the system can be used at any time
2. The use of Single Compression Balance Spring device, good regulation, no noise, indefatigable, not easily broken, safe and reliable.
3. A variety of input and output modes can be selected, normally open normally closed are optional.
4. Speed can be adjusted.(1s-4s)
5. Barrier gate up &down signal relay switch output.

6 R&G Traffic light relay switch signal output.

7. External loop detector signal anti-smashing interface, infrared sensor signal anti-smashing interface.
8. Count mode interface.
9. Delay closed selection function.
10. Parking system interface.
11. High-strength precision cast steel, industry moving parts beyond the industry quality

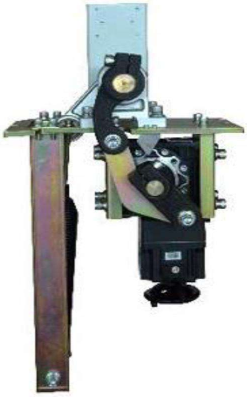
4. Technical Parameter List

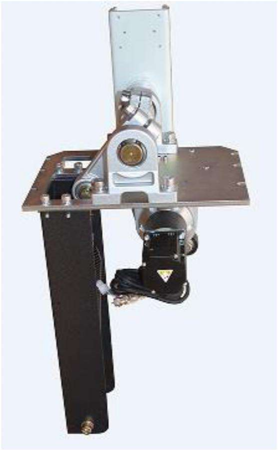
33T-BLDCW barrier gate	**V15T**	**V30T**
Max Boom Length and Diameter	3m~4m, Φ75mm,80*45mm	4m~6m, Φ75mm, 80*45mm
Open/Close Time	1.5sec~2s	2.5sec-4s
Power Supply	24/100/240VAC,50/60Hz	24/100/240VAC,50/60Hz
Power Consumption	max.140W	140w
Machine Case	Varnished 1.8mm steel plate	Varnished 1.8mm steel plate
Machine Case Size(W*D*H)	320*320*960mm	320*320*960mm
Net Weight	45kg	45kg
Operating Temperature Range°C	-25℃~+80℃	-25℃~+80℃
Driving Method	24V BLDC motor	24V BLDC motor
Protection System(IP)	54	54
Insulation Class	F	F
Relative Humidity	≤90%	≤90%
Motor No-load Speed	3000r/min	3000r/min
Remote Control Distance	≤100m (open, sunny weather)	≤100m (open, sunny weather)

Output Torque	1.2N.M*140=160N.M	1.2N.M*140=160N.M
Running Noise＜	45dB	45dB
Surface Treatment	Outdoor metal powder electrostatic spraying	Outdoor metal powder electrostatic spraying
DZ Controller		
Voltage(±10%)	100-240 volt	100-240 volt
Frequency(Hz)	50-60	50-60
Max Power Consumption(watt)	5	5
Fuse	F5AH	F5AH
Control Voltage	24V DC/max.3000mA	24V DC/max.3000mA
Input-end Number	5	5
Open Collector Output Number	2	2
Zero Level Reversal Relay Number	2	2
Operating Temperature Range℃	-40℃~+85℃	-40℃~+85℃
Protection System(IP)	20	20
Weight(gram)	450gram	450gram
Dimension	mm(L-W-H)	181*121*50

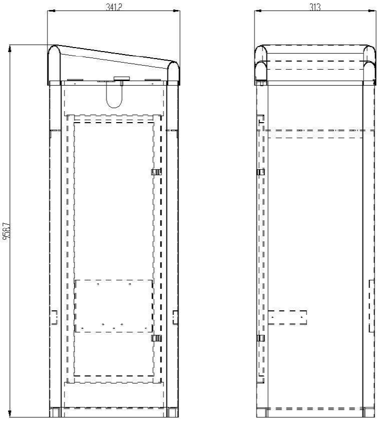
5.Machine Core Components

Worm gear reducer movement



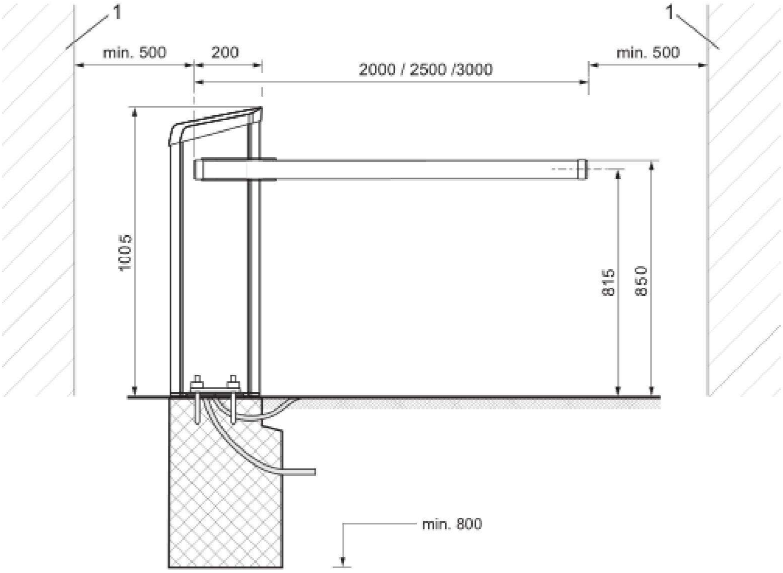


Planetary gear reducer movement

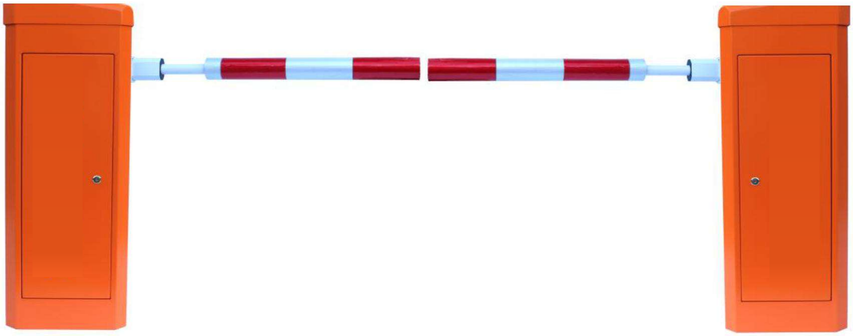


Machine Case Size

6. Product Dimension and Installation Direction definition



Installation direction definition:



(Left-side fixed,**Right Version**) (Right-side fixed,**Left Version**)

Frontside of gate Frontside of gate

7.Security Tips

7.1 General Security Information

The design, manufacture and testing of the barrier system are using of the "state-of-the-art" technology and can only be shipped after the satisfaction of the strict safety and reliability of technical standards. Even so, if the barrier system is not properly operated, it will also cause danger to people and property. So this manual must be fully read and all of the safety information must be followed.

If the barrier system is improperly used or used for non-intended use, the manufacturer will refuse to be responsible for the resulting consequences and will cancel the original quality assurance.

7.2 Suggested Application

High-speed barrier gates can only be used for toll stations for highways and main roads.

The medium speed barrier gates are designed to control the entrance and exit channels for parking area, multi-storey car parking lot and other vehicular access routes.

The controller is specifically designed to control the barrier gates.

It is not permissible to use these barrier gates for any other purpose.

Any changes and replacements to the barrier gates and the control components are prohibited.

7.3 Safety sign

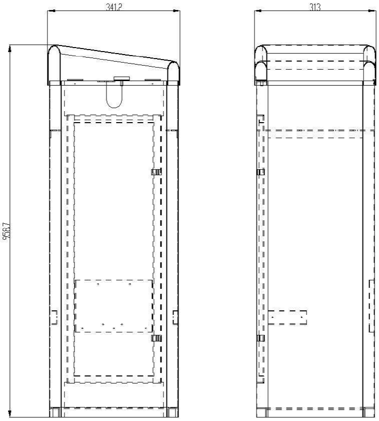
The following symbols used in this instruction are used to indicate potential hazards and other safety information.

⚠ Be careful!

Use this LOGO to indicate these actions or situations that the property and equipment are in danger.

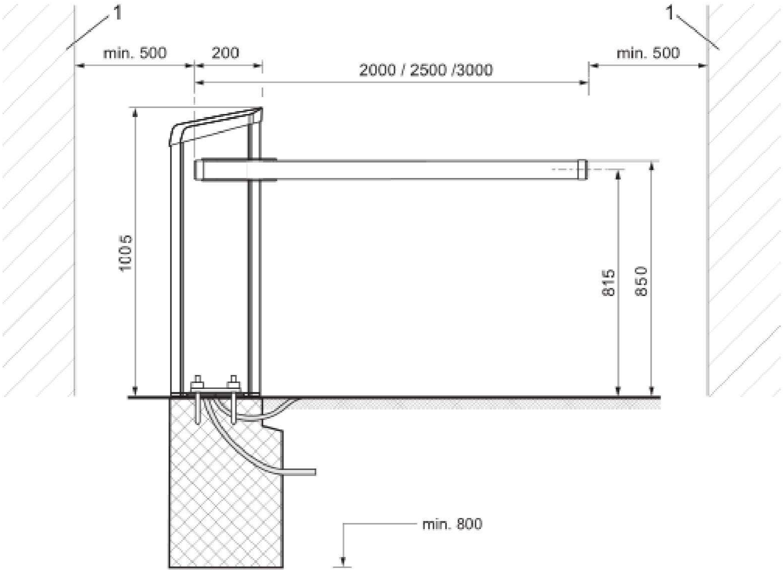
Watch out!

Use this LOGO to indicate the useful information for the operator



Machine Case Size

6. Product Dimension and Installation Direction definition



Installation direction definition:

7.4 Operating Safety

A safety distance of at least 500 mm must be provided between the end of barrier gates and the closest physical obstructions (such as buildings, walls and fences, etc.). The activities components of barrier gates must be able to be seen directly by operators. When the boom is moving, the personnel and other objects are forbidden to stay near the barrier gate. If you have installed the barrier and operating elements and are connected to a fixed main power supply, you must use a main circuit switch with full polarity insurance device. It must be operated in accordance with this installation manual, and must be approved by the company before making any changes. The fixing device of the boom is designed to bear winds of magnitude 10 (= 500N / ㎡.). If it's expected that there will be a higher speed wind, must be approved by the company before change the fixing device of boom. A boom length more than 3.5meter will be required to install a swing support or a support column. All of the circuit connections, wiring work and replacement of any circuit components must be carried out by professionally trained electricians. Please firstly cutting off the connections with main power supply before opening barrier gates. It is forbidden to make technical modification or changes to the barrier system.

After cutting off the power, please starting after at least 10 seconds.

7.5 Technology Development

The manufacturer reserves the right to modify the technical specifications of the product to meet the latest technological developments without prior notice. The company is glad to provide the information about current operating manual and any information about the changes and extensions.

8. Assembly and Installation

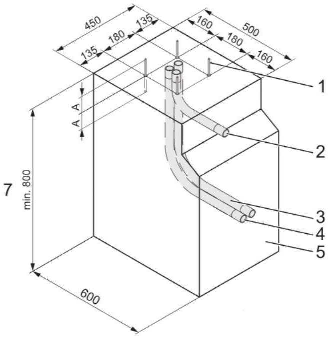
8.1 Installing Basement

To ensure these devices are firm and stable under all operating conditions, it is necessary to provide a foundation as the following dimensions:

Basement depth: at least 800mm (without the frost)

Basement basal area: 500(length)*600mm(width)

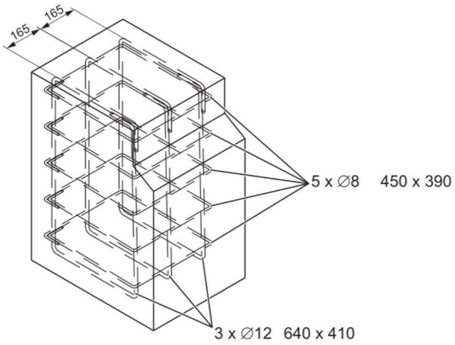
In the above section of the foundation that face to the side of the vehicle channel, is more 100mm than other three side (pictures as below:).



1. Fixing bolts (4 pcs)
2. 29mm diameter cable tube for inductive coil wire
3. 29mm diameter cable tube for power wire
4. 29mm diameter cable tube for controller wire
5. Concrete basement (BH PC250, strength = 25N / mm2)

It must embed a plurality of hollow pipes with a diameter of 29mm in the foundation to accommodate the main power supply wire, the control signal wire and the possible inductive coil wire (that is, two or three hollow pipes for the barrier;2 hollow pipes for the control column). It is absolutely necessary for a reinforced rebar structure to ensure the firmness of the concrete basement. (picture as below:)

Reinforced structure for concrete foundation



The concrete in the foundation is BH PC 250 grade concrete (strength 25N/㎡).

The base surface must be smooth so that the barrier machine case can be level and horizontally installed on it. After the concrete has been poured, the fixing screw holes can be drilled on the base surface according to the position shown on picture.

Drilling distance: 170mm (square arrangement, see picture)

Hole diameter: 16mm Hole depth: 100mm

After drilling the hole, install the four M16 × 150 set fixed screws in four holes.

5×Ø8 440x440 St 111
3 ×Ø12 740x410 St 111

8.2 Basic Mechanical Structure

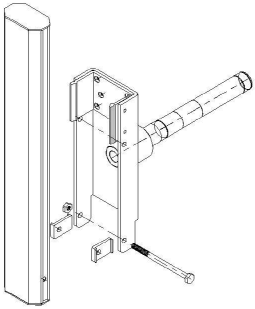
When the spring is tightened, the spring and the brake arm exert a considerable force on the drive and thus creating a potential risk of injury.

So any work on the boom drive must be carried out when the spring is not tightened and the boom is safe or discharged.

Boom installation:

A: put the boom into boom tray head according to the hole location.

B: use the spanner to clamp the round boom by screws, gasket and screws nut



8.3 Springs Testing and Adjustment

All of the barrier gates already set up in the factory based on the original boom. The spring inside of the mechanism is in a highly tense status. And all the adjustment of the barrier must be carried on after the spring is in loose status and the boom is put aside. But when you use it, please check the boom's setting after installed it and on the first operation. After the weight of boom is balanced by the tension applied by the spring, the barrier gate can be operated correctly. Therefore, any changes to the boom must be re-adjusted according to the following steps.

Testing spring setting:

1. open the door of barrier machine case with a key and remove the mounting plate.
2. unplug the power cord
3. manually adjust the boom to the position of about 30 degree and release the hand. If the boom is stable in this position and not move, indicating that the spring adjustment is correct.

Adjustment spring setting:

1.According to the different boom length to assemble the compression spring, tighten or loosen the adjusted handle to make the boom maintain stable on 45° , this is the best spring strength. Until the boom keep stable on the position of 45 ° angle.

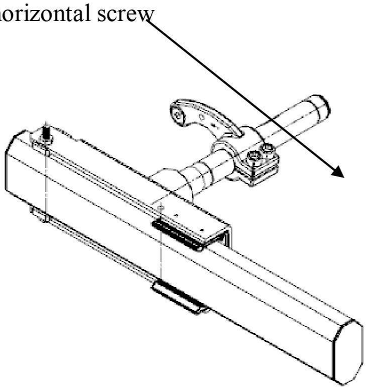
Exceptions:

When the barrier gates are set to automatically open when power failure, a greater spring force is required than the above mentioned (only when the max boom length is 3.5)

8.4 Adjust the Position of Barrier Arm

To adjust the position of the boom (for example, after exerting excessive force), please take the steps as below:

1. open the barrier gate door and remove the cover
2. loosen the two fasten screws of the DZ-1 on the boom shaft with the M12's Allen wrench, so that the boom can be repositioned by hand.
3. adjust the position of the boom (horizontal position, as shown in picture 1).
4. Use the torque wrench to tighten the two fastening screws (72 Nm)



Adjust the horizontal screw

(picture 1)

8.5 Manually Unlocked Open/Close Arm When Power Off

The barrier gate may into the vertical or horizontal mechanical dead point position when power off, if manually up/down the arm at this time, please follow the steps as below:

make the key rotate clock wise 90 degrees and open the door

unscrew the two butterfly nuts upward, open the cover

lifting the boom or dropping the boom: Use a hand to turn the handwheel, the hand wheel below the motor, which can lifting or dropoing the boom

9. Power Connection

Connect the L,N to the air switch and P E connect to the ground screws (yellow-green line)

Important! Can be only use the randomly provided power connection cables to connect the main power supply (internal power has already connected in the factory)

Can't connect the AC power to the controller.

Attention! After cut the power off, start the power supply prematurely will

bring the damage to the equipment.

This is an optional accessory, after use, it can support automatic opening when power is off.

10. DZ BLDC Control Board

⚠ Warning!

DZ BLDC controller is specially designed for the use of high speed, medium speed barrier gates. The configuration for most of the barrier gate systems can be accomplished by using a standard version controller. The position of the boom is continuously detected by the sensor. It replaces the limit switches used in conventional barrier control systems. The combination of the sensor and the controller unit possibly ensures the optimum control to the boom running.

Attention!

If some special features are included, the controller's connection may differ from the situation shown in the illustration. Please refer to the additional line design in the envelope.

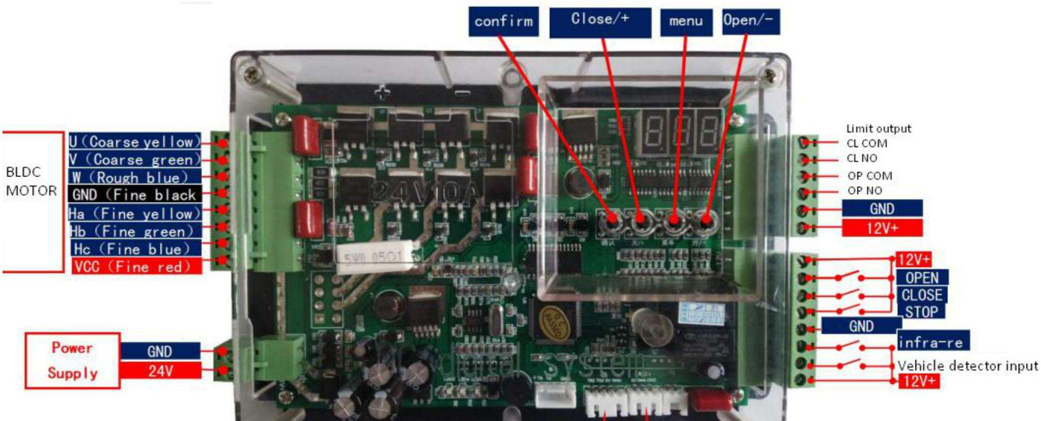
Before power on, please check the following items:

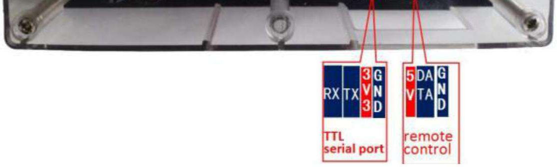
1. The power supply is 24 V 10 a DC power supply.
2. Whether the polarity and sequence of all external wiring (power line, motor line, ground sensing line) are correct, and the connection is firm and the wiring is stable.

10.1 Command Generator: connection and functions in different mode

☞ **Attentions:** All command generators (pushbuttons, limit switches, etc.) must be connected by the volt-free contact way. Please consult the company before connecting the command generator and the output signal. The wiring panel shown as bellows.

The control box has two card slots, a fixed screw, in order to conveniently wiring, unscrew the fixed screw to take control box quickly remove from the card slot to connect wire





Left side port is to connect to BLDC motor and power supply. The motor interface is: Vcc +5V red line, Hc blue (small), Hb white or green(small), Ha yellow (small),

GND black, W blue (Big), V white or green (Big), U yellow (Big). Right side port is connect to control system. And position signal output.

1. After the gate is powered on, the digital tube of the main board displays "190". For the first self-test of the main board and the motor, it is necessary to manually press the "on" and "off" buttons on the main board to conduct self-test and learn the stroke once. The main board automatically memorizes the motor travel. After the subsequent power failure, only the remote control or the license plate recognition camera can send the "on" or "off" instructions.

2. The self-test of the barrier gate is two limit positions fixed by the learning motor. Please ensure that when the button self checks the learning stroke, the opening position is greater than 90 ° and the closing position is less than 0 ° so as to facilitate the subsequent horizontal and vertical debugging.

Note: after the first installation and power on for the first time, you must use the "on" and "off" buttons on the main board to complete self-test and learn the opening and closing stroke once!

Please pay attention to the following items during self inspection:

1. Whether the motor polarity is correct: if the motor polarity is wrong, the self-test of the barrier gate cannot be completed.

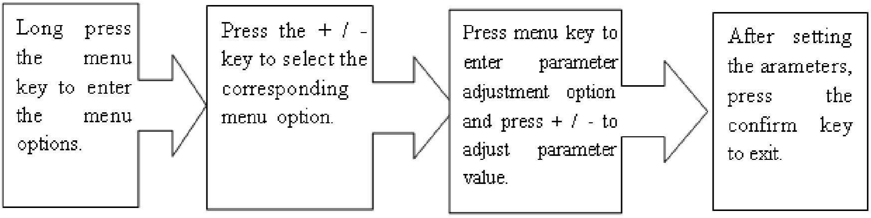
The motor polarity adjustment menu is L-d

2. Whether the left and right directions are correct: if not, when the motor is "on", the gate actually acts as "close". When the motor is "off", the gate is actually "on".

The digital state displayed by the main board nixie tube will be reversed accordingly,

The normal is: "on", the number of digital tube increases, and "off", the number of digital tube decreases. If not, adjust the menu L-d.

Operation process:



Long press the menu key to enter the menu options.

Press the + / - key to select the corresponding menu option.

Press menu key to enter parameter adjustment option and press + / - to adjust parameter value.

After setting the arameters, press the confirm key to exit.

function	letter menu	number menu	Parameter range	Default value	explain
Opening speed regulation	L-1	L01	20-95	65	The larger the value is, the faster the opening speed is; the smaller the value is, the slower the opening speed is.
Closing speed regulation	L-2	L02	20-95	65	The larger the value is, the faster the closing speed is, and the smaller the value is, the slower the closing speed is.
Adjustment of stability of brake rod in place	L-3	L03	8-15	12	The smaller the theory is, the more stable it will be, but if it is too small, the gate lever may not reach the position.
Response time sensitivity adjustment of anti crashing vehicle	L-4	L04	15-30	30	Theoretically, the smaller the gate is, the more sensitive it will be. However, if it is too small, it will automatically lift the lever when it is half way down.
Motor force value adjustment/ Menu display form setting	L-5	L05	70-90	70	When the numerical value is large, the motor force is large, the value is small, and the motor force is small, When the setting parameter is even, the nixie tube is displayed as an alphabetic menu, When the number is odd, the digital tube displays the digital menu.
Horizontal position adjustment of gate lever	L-6	L06	3-90	20	The smaller the value is, the smaller the angle of gate rod is.
Manual / automatic mode selection	L-7	L07	0-6	0	0 (manual) 1 (automatic) The following 1-5 are the interval time of automatic operation, 1 is the fastest and 5 is the slowest 6 is the half stroke operation of the gate.
Opening memory function setting.	L-8	L08	0-1-4	0	0 (without memory) 1 (with memory) 4 (peak mode, after opening the gate in place, Press the stop button of the remote control to turn on the function. To cancel, press the switch off button of the remote control).
Vertical position adjustment of gate lever	L-9	L09	3-90	20	The smaller the value is, the smaller the opening angle of gate rod is.
Adjustment of deceleration stroke of the first section	L-L	L10	20-90	55	The smaller the value is, the smaller the starting and decelerating travel is.
Adjustment of deceleration travel of first section closing railing	L-b	L11	20-90	55	The smaller the value, the smaller the off deceleration travel

Anti smashing force adjustment	L-C	L12	15-100	102	The larger the value is, the greater the anti smashing force will be. If the value exceeds 100, the anti smashing function will be cancelled, The default setting of the factory is 102, which cancels the anti smashing function, and the customer needs to adjust the parameters.
positive and negative operation of	L-d	L13	0-3	1	1: Motor negative, reducer forward 2: Motor negative, reducer reverse
Strength value of starting motor of barrier gate	L-E	L14	20-40	30	Motor running speed during self inspection
Add remote control	L-F	L15	0-255	0	Enter the menu, 000 is the increase state, 253 automatically clear the remote control and then exit automatically
The second paragraph Adjustment of brake opening / closing deceleration travel	L-H	L16	1-10	0	Use of 0.3 second Expressway barrier
Power off automatic pole lifting function	L-p	L17	0-21	0	The parameter value represents the voltage value, which is generally set to 20 or 21. When the voltage is lower than the set value, the switch will be opened automatically. (this function can only work with a backup power supply installed)
Traffic light / light strip / detection state mode adjustment	L18	L18	0-2	0	0 is traffic light mode, 1 is in place detection mode, 2 is tricolor light strip Mode (need to be realized by external relay)
Adjustment of earth induction delay and gate closing time	L19	L19	1-255	000	The setting value represents the delay of rod dropping time for several seconds after the earth pressure sensing is completed. For example, 1 is 1 second, 2 is 2 seconds, and 000 represents that this function is disabled.
Adjustment of automatic rod falling time for lifting limit	L20	L20	1-255	000	Set the value, which represents the delay of rod falling time and waiting for several seconds, for example, 1 is 1 second, 2 is 2 seconds, 000 means this function is disabled.

Spring adjustment:

When changing the size of the Boom length, the spring force must be adjusted to match the length/weight of the Boom.

Before using the barrier gate, please check the spring first and please follow the following steps.

1. Open the housing door by key;
2. Unplug the power;
3. loosen the two fasten screws of the DZ-1 on the boom shaft with the M12's Allen wrench, so that the boom can be repositioned by hand.(see the picture1 on 8.4)
4. Put the boom in an angle of 30 degree by hand; then release the boom. If the boom does not move, it means it is well-balanced. And the barrier gate can run normally. If the boom doesn't stay and goes up start, then the spring force is too high and needs to be reduced. If the boom is falling down, then the spring force is too low and you need to increase the spring force.
5. If is not balanced, please release the fixing bolt of the spring. The user can loosen or tighten the spring until the boom stays still in an angle of 45 degree.
6. Finally restore to the vertical state after torque screw

The following is the corresponding controller parameter adjustment for different Boom lengths，

Used for Worm gear reducer

	Worm gear parameters			
Para-meters	3m	4m	5m	6m
L-1	85	80	70	60
L-2	85	80	70	60
L-3	12	12	12	12
L-4	30	30	30	30
L-5	70	70	70	70
L-6	10	10	10	10
L-7	0	0	0	0
L-8	0	0	0	0
L-9	10	10	10	10
L-L	45	60	70	80
L-b	45	60	70	80
L-c	40	40	40	40
L-d	1 or 3	1 or 3	1 or 3	1 or 3
L-E	25	25	25	25
L-F	0	0	0	0
L-H	0	0	0	0
L-P	0	0	0	0
L-18	0	0	0	0
L-19	0	0	0	0
L-20	0	0	0	0
L-21	1	1	1	1
L-22	30	30	30	30

Used for planetary gear reducer:

	Planetary reducer parameters			
Para-meters	3m	4m	5m	6m
L-1	60	50	35	25

L-2	60	50	30	20
L-3	10	10	10	10
L-4	30	30	30	30
L-5	70	70	70	70
L-6	3	3	3	3
L-7	0	0	0	0
L-8	0	0	0	0
L-9	5	5	5	5
L-L	70	80	80	120
L-b	70	80	80	100
L-c	20	20	20	20
L-d	Left Version is set 1,Right Version is set 2			
L-E	20	20	20	20
L-F	0	0	0	0
L-H	0	0	0	0
L-P	0	0	0	0
L-18	0	0	0	0
L-19	0	0	0	0
L-20	0	0	0	0
L-21	1	1	1	1
L-22	30	30	30	30

11.Technical Support

If the operational failure can’t be handled by your technical staff, please contact our authorized service representative or professional assistance.

Please provide the barrier gate model, product serial number, version and other information when contact us for technical support services, which you can find in barrier gate model plate.

12. Packing List

Title	Specification	Qty	Unit	Note
Barrier gate boom		1	Piece	
Machine case press board		2	piece	For fixing case
Expansion bolt	M16X150	4	set	For fixing case
Supporting pole		1	piece	Optional part
Expansion bolt	M8	4	unit	For fixing boom
Boom fixed screws	M12*70	2	unit	
Case Key		2	unit	
Balance spring		1	unit	For spare usage
Remote controller		2	unit	
Machine control		1	unit	Optional
Instruction		1	piece	

13. Service Items

1. Two year's free servicing is supplied (not including the arm and remote controller). In the condition of complying with the operating instruction, without the unauthorized maintenance of the parts and the machine have no obvious mechanical damage, we will provide a quality assurance for the barrier gate product from the date of use for a period of two years or from the date of shipment up to three years including all mechanical and electronic components other than special consumables.
2.Lifetime charged servicing is offered.
3.Technology servicing is supplied.
*The following situations are charged for servicing (or replacement):
1.Broken by wrong installation;
2.Broken by improper voltage;
3.The surfaces of the system destroyed by wrong installation or use.
4.Broken by natural disaster.
5.Overdue.
6.Servicing items out of the promises.

Remote Control Learning Code Instruction

Remote Control Learning Code Instruction

Add remote control

Remote control pairing: enter the mainboard menu to L-F [L15], and the LED displays "000". At this time, press any button on the remote control, and hear the main board drip, and automatically return to the menu selection status, indicating that the pairing is successful. Then press the save exit key to exit the menu.

Clear remote control: clear all paired remote controls, enter the main board menu to L-F [L15], press the menu key to enter the learning state, adjust the parameter to 253, and automatically return to L-F [L15], at the same time, all remote controls are cleared。

Note: before learning the remote controller for the first time, clear all remote controllers in the motherboard first, so as to avoid the existence of other remote control codes, which may affect the normal use.